(Please write your Enrollment Number)

Enrollment No._____

MID-TERM EXAMINATION

(Course Name: B. Tech. ECE-AI-1 & ECE-AI-2) (Semester :3rd)

(October 2024) OFF LINE mode

Subject Code: BEC-211	Subject: COMMUNICATION SYSTEM
Time: 1 ½ Hours	Maximum Marks: 30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)	CO Mapping
	a) Draw the Frequency and Time domain representation of AM.		
	b) Define Modulation and also describe the need of modulation in brief.		
	c) Define the statistical averages of Random Process with its expressions.		
	d) Calculate the percentage power saving when the carrier and one of the sidebands are suppressed in an AM wave modulated to a depth of (a) 100 percent & (b) 50 percent.		
	UNIT I		CO Mapping
Q2	**Attempt any two parts:**	(5*2=10)	
	a) Define Random Process. Differentiate between Strict-sense stationary (SSS) and Wide-sense stationary (WSS) random processes.		
	b) A continuous Random Variable is uniformly distributed in the interval (-10, 10). Find all of its statistical averages.		
	c) Plot the Probability Mass Function for a random variable which is specifying number of heads in the experiment of tossing a coin twice.		
	UNIT II		CO Mapping
Q3	**Attempt any two parts:**	(5*2=10)	
	a) Derive the expression for Power Content of Single Tone AM. Compare SSB-SC, DSB-SC and VSB.		
	b) What are the different detectors of AM. Explain with the help of circuit diagram Envelope Detection Scheme.		
	c) Explain Ring Modulator with the help of neat diagram.		